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Real-Time Digital Twin–Driven 3D Near-Miss Detection System at Construction Sites

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Abstract: To enhance on-site safety management, detecting near-misses in a timely and accurate manner is crucial. However, traditional monitoring methods that rely on inspectors are time-consuming, labor-intensive, and limited in their ability to detect three-dimensional (3D) incidents. Digital twin technology presents a promising solution by integrating real-time monitoring, simulation and interactivity capabilities in three dimensional virtual environments. Therefore, we propose a real-time digital twin–driven near-miss detection system to facilitate safety management at construction sites. Using advanced stereovision techniques, the system reproduces the construction site as a digital twin model, enabling the real-time 3D monitoring of near-miss events. Additionally, the system incorporates an early warning mechanism to alert workers and managers of potential safety hazards. To demonstrate the effectiveness of the proposed system, a lifting project under various exposure conditions was investigated. The results demonstrate the system's ability to detect and visualize near-misses in three-dimensional space and provide proactive warnings of potential hazards. **DOI: 10.1061/JCEMD4.COENG-15583.** *© 2025 American Society of Civil Engineers.*

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Introduction

Safety is a crucial concern in the construction industry, because lack of safety leads to fatalities, injuries, and accidents that result in substantial costs and irreversible damages every year (Rokooei et al. 2023). A comprehensive investigation of numerous construction safety accidents found that almost every occurrence of a safety accident is not a coincidence without premonitory signs (Shen and Marks 2016). Construction sites can be high-risk areas for serious safety accidents due to the accumulation of various incidents influenced by hazardous factors associated with building processes. However, in certain situations, these risk factors can be mitigated or eliminated, preventing any actual harm or significant damage (Khakzad et al. 2014). Thus, only minor injuries and minimal losses may occur (Zhang et al. 2019), leading to what is known as near-miss events at construction sites (Raviv et al. 2017). The Occupational Safety and Health Administration (OSHA) defined a near miss as a critical incident that narrowly averted an accident due to timely intervention (OSHA 2016).

Heinrich's law (Heinrich 1941) suggests that for every major accident, there are typically 29 minor accidents and 300 potential hazards. This implies that injuries or fatalities are not isolated incidents, but rather the result of a series of events occurring one after another, even if the incident itself happens suddenly at a specific moment (Fig. 1). Near-miss events serve as precursors to accidents, highlighting the importance of paying attention to these near-misses in order to prevent safety incidents (Saleh et al. 2013). Consequently, the real-time and effective detection and analysis of near-miss events can reduce the frequency of safety accidents and enhance the overall safety management at construction sites (Duan and Zhou 2023).

However, the real-time detection of near-miss events at construction sites poses a complex and formidable challenge (Goldenhar et al. 2003; Hajdasz 2015; Yang and Cai 2024), highlighting the dynamic and unpredictable nature of construction environments. The traditional approach to on-site safety inspections, which is heavily reliant on professional surveyors, demands significant workerpower and material resources. Moreover, manual monitoring is prone to errors and is less reliable due to factors such as staff fatigue or prolonged strain on vision (Cheng et al. 2024), resulting in an underestimation of risk factors at the construction sites.

To address these issues, various solutions have been proposed in both academia and industrial settings. An innovative safety model employing the Internet of Things (IoT) has been designed to detect near-miss events in real-time through vigilant monitoring of site personnel and environmental conditions (Chung et al. 2023), with the goal of reducing accident rates. Recently, blockchain technology has emerged as a cutting-edge tool for monitoring near-misses at construction sites (Li et al. 2023) laying the groundwork for an extensive database of incident records and an early warning system for near-miss detection. Additionally, 2.4-GHz Bluetooth Low Energy (BLE) technology has been utilized to develop a real-time location system to accurately gather and tally near-misses at construction sites (Li et al. 2016). However, contemporary research on near-misses detection predominantly has focused on two-dimensional analysis, and the transition to more-comprehensive three-dimensional (3D) tracking is still pending. This is due to

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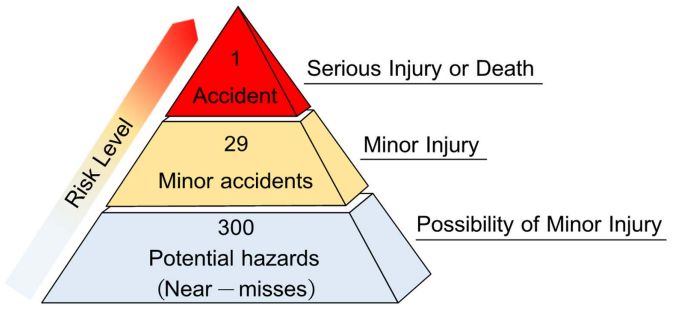


Fig. 1. Diagram of Heinrich's law.

largely the intricate nature of construction site environments and the perpetual movement of personnel and machinery, which complicates data collection and analysis. In addition, some near-misses that occur cannot be observed intuitively.

In recent years, the construction industry has begun exploring various new technologies, including digital twin (DT) technology. This technology enables the real-time collection of construction-related information, such as workers, materials, equipment, and so forth, which is displayed intuitively on a general platform. Individuals at different phases of the construction process can access this information for strategic planning in their next steps. Moreover, digital twin technology plays a vital role throughout the entire life cycle of a building, encompassing the design, construction, and operation and maintenance phases. To optimize the planning scheduling and execution of prefabricated components at construction sites, a real-time synchronous digital twin system has been proposed (Jiang et al. 2022c). In addition, digital twins also have been applied to anomaly detection and monitoring during the operation and maintenance phases of construction projects (Lu et al. 2020). Another notable application is the use of a vision-based framework for digital twinning and threat assessment, which identifies and evaluates the characteristics and potential impacts of windborne debris within a virtual simulation model of a construction site (Kamari and Ham 2022). However, current research has not yet explored the application of digital twins for the detection of near-misses at construction sites.

Therefore, this paper proposes a digital twin–driven near-miss detection system at construction sites in a timely manner. The remaining parts of this paper are organized as follows: section "Related Works" presents research on near-misses and digital twin technology within the construction industry. Section "Proposed Digital Twin–Driven 3D Near-Miss Detection System" outlines the architecture of the proposed digital twin–driven near-miss detection system and details each module of the framework. Section "Pilot Study" presents the on-site validation and robustness testing of the system at a construction site. Section "Discussion" analyzes and discusses the outcomes of the implementation. Finally, section "Conclusions and Future Work" summarizes the study, highlighting its contributions to the field and suggesting avenues for future research.

Related Works

Near-Misses in Construction

Although incidents may pose risks of injury or death, they can be prevented with proactive measures (Saurin et al. 2015). Despite the importance of gathering detailed data on near-miss incidents for construction safety management, too few of these events are identified and reported. At present, the existing research on near-misses in the construction field is divided into two categories.

One category traces the roots of near-misses at construction sites. Wu et al. (2010a) established a systematic approach to address and prevent precursors and immediate factors (PaIFs) at construction sites, enabling the extraction of valuable insights from accident histories to enhance site safety. Zhou et al. (2019) introduced a human dynamics model using attempt data to uncover the shared dynamic characteristics of accidents stemming from unsafe behaviors among construction workers. Li and Ding (2019) proposed a deep learning-based video surveillance method for automatically detecting and analyzing the possible causes of near-miss events at construction sites, forming an exhaustive case base from real-time monitoring for safety assessment, training, and safety management.

Another category monitors near-miss events for analysis to guide construction. Wu et al. (2010b) proposed an autonomous real-time tracking system of near-miss accidents (ARTTS-NMA) at construction sites to monitor near-misses. Yang et al. (2014) introduced an inertial measurement unit (IMU)-based method for detecting attempted falls, which provided quantitative data for fall risk assessment of steel workers. Teizer and Cheng (2015) proposed a proximity hazard indicator to automatically identify hazardous areas at construction sites and to collect and analyze spatiotemporal conflicts between walking workers and identified hazards.

The evolution of near-miss detection technologies has been marked by advancements in wearable devices, mobile technology, and artificial intelligence. Early efforts utilized wearable inertial measurement units to monitor workers' movements and detect potential falls (Aria et al. 2014). Subsequent research refined these systems with automated detection methods and threshold-based algorithms (Yang et al. 2014, 2015, 2016).

Mobile technology introduced a new paradigm, leveraging smartphone sensors and artificial neural networks (ANNs) for event recognition, capitalizing on the ubiquity of smartphones (Zhang et al. 2019). However, this approach faced limitations in battery life and computational power. Proximity-based systems used Global Positioning System (GPS) and radio-frequency identification (RFID) technologies to assess risks associated with hazardous proximity, providing valuable insights into potential near-miss incidents (Golovina et al. 2016). Real-time location tracking with ultrawideband (UWB) systems further enhanced this approach by offering precise tracking of workers' movements in relation to hazardous zones or equipment (Teizer and Cheng 2015). However, these methods are hindered by high costs and installation complexities.

Despite progress, current technologies face limitations including discomfort and usability issues with wearables, battery and processing constraints with mobile devices, and cost and complexity challenges with proximity-based systems. To address these limitations, computer vision offers a promising alternative. By analyzing video feeds from cameras strategically placed on construction sites, computer vision systems can detect near-miss incidents with high accuracy and minimal user discomfort. This approach not only reduces the reliance on wearable devices and mobile technology but also provides a more comprehensive view of the construction environment, enhancing the overall safety monitoring system.

Application of DT in Construction

In recent times, the concept of digital twin has been extensively defined and analyzed by numerous scholars (Keskin et al. 2022). Akanmu et al. (2014) emphasized that DT fundamentally links a physical object with its digital counterpart. Expanding on this

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notion, Tao et al. (2019) suggested that a DT should encompass five integral components: the physical entity, its virtual representation, the interconnecting networks, the underlying data, and the associated services. Jiang et al. (2021) succinctly captured the essence of DT and identified its key elements within the realm of civil engineering.

With the development of technologies such as the Internet of Things and computer vision, the concept of digital twins has been applied throughout the life cycle of projects (Cimino et al. 2019). During the construction phase, digital technologies facilitate the real-time monitoring and management of construction-related information, encompassing processes, machinery, materials, workforce, quality, and safety at the job sites (Jiang et al. 2022b). For example, Pan and Zhang (2021) suggested a comprehensive digital twin framework for managing projects. Jiang et al. (2022a) created a digital twin framework for construction activities that involve human–machine interaction, focusing on construction hoisting, and analyzed the variation patterns of tower crane stability with incremental loads. Hou et al. (2020) noted that although digital twin technology has been shown to improve construction worker safety, the industry has yet to fully leverage these benefits in practical applications. Currently, the application of digital twin methods at construction sites is still limited. Yu et al. (2021) highlighted that implementing the digital twin approach largely depends on data, particularly the twin data derived from physical entities, virtual models, and relevant domain knowledge, which must be integrated into each component. However, the constantly changing construction personnel, dynamic construction machinery, and complex construction environment pose challenges in acquiring multisource information at construction sites. Additionally, there is a lack of interaction between the data, such as the interaction between workers and buildings, construction vehicles, and so forth, which hinders the real-time and accurate management of construction site safety.

In summary, DT holds significant potential for real-time monitoring of near-miss events at construction sites. Firstly, DT can provide a three-dimensional representation of near-miss incidents by mirroring the physical environment in a virtual space, allowing for the capture of detailed and accurate data that might otherwise be missed. This capability enables the identification of subtle anomalies or deviations from safe practices that could lead to accidents. Secondly, by leveraging advanced analytics and machine learning algorithms, DT can process this real-time data to perform predictive analysis, issuing early warnings of potential safety risks before they escalate into actual accidents. Furthermore, DT supports low-cost remote simulation, computation, analysis, optimization, and management processes (Long et al. 2024), thereby enhancing operational efficiency and safety management across construction sites. However, few studies have demonstrated the application of digital twin technology in the field of near-misses detection at construction sites, and its effectiveness needs to be explored. Therefore, based on the concept of digital twin, this study utilized computer vision technology to integrate multisource information of the construction fields and establish a near-miss detection system for construction sites.

Proposed Digital Twin–Driven 3D Near-Miss Detection System

Classification of Near-Misses at Construction Sites

To provide a more comprehensive analysis of site operations and interactions, we referred to relevant literature (Li et al. 2016) on near-miss accidents and, based on this, combined with discussions with on-site construction management personnel, propose the classification principles of near-miss accidents. The paper categorizes the construction site into two distinct groups: static elements, which include structures and construction materials that experience minimal changes over short periods, often displaying cyclical patterns during the construction phase, typically over the span of a week to 2 weeks; and dynamic elements, which are entities that change in real-time, such as workers, materials being transported, and construction machinery.

Correspondingly, hazardous regions at the construction site are categorized into two categories: static fixed areas, in which hazards result from interactions between individuals and static materials or structures; and dynamic changing areas, in which hazards arise from interactions between individuals and dynamic materials or construction machinery. Moreover, the hazard levels are divided into three categories: safe, in which workers are not in any danger in this area; hazardous, in which where workers may be in danger in this area; and admonitory, which is set around determined hazardous areas.

There are two types of near-misses in static fixed areas. The first type exhibits diffusive characteristics, and is denoted S1 [Fig. 2(a)]. A typical example is a temporarily excavated foundation pit, in which the boundary of the pit defines the hazardous zone. The fencing around the pit, designed to keep workers at a safe distance, defines the safe area. The region between the fence and the pit's edge serves as an admonitory area. Another common example is an aerial work platform or scaffolding, in which the area occupied by workers is considered safe. Workers receive warnings when they approach the edge, and a dangerous area lies beyond the edge. These near-misses demonstrate anti-diffusion-like characteristics, and are denoted S2 [Fig. 2(b)].

There also are two types of near-misses in dynamically changing areas. One involves horizontal hazards, and is denoted S3 [Fig. 2(c)]. When construction machinery, such as vehicles and cranes, is in motion, workers may face accidental injuries resulting from falls and collisions. Given the mobility of both workers and hazard sources, the dynamically uncertain hazard source area is defined by establishing the maximum effective area boundary within a fixed distance. Conversely, the other type of danger arises from the vertical direction, such as various construction materials being hoisted on site, in which workers may face the danger of falling objects from high altitudes, and is denoted S4. Therefore, the space below the suspended object is designated as a hazardous area [Fig. 2(d)].

Design Framework of the System

To monitor the four types of near-misses in real time, especially on complex and chaotic construction sites, the system in this paper was built based on the multisource information of construction sites, including buildings, construction personnel, construction materials and construction machinery and equipment at the construction sites. Considering the importance of near-misses in construction site safety management (Jin et al. 2020), digital-twin technology is introduced to the field of construction site near-misses detection in this paper.

The digital twin concept, comprising five components—physical part, virtual part, connections, data, and services (Tao et al. 2019)—is illustrated in the system design framework presented in Fig. 3. The platform operates as follows. First, images of the construction site, which are transmitted to the image processing model including target recognition, tracking, and stereo matching algorithms, are collected in real time using drones and binocular

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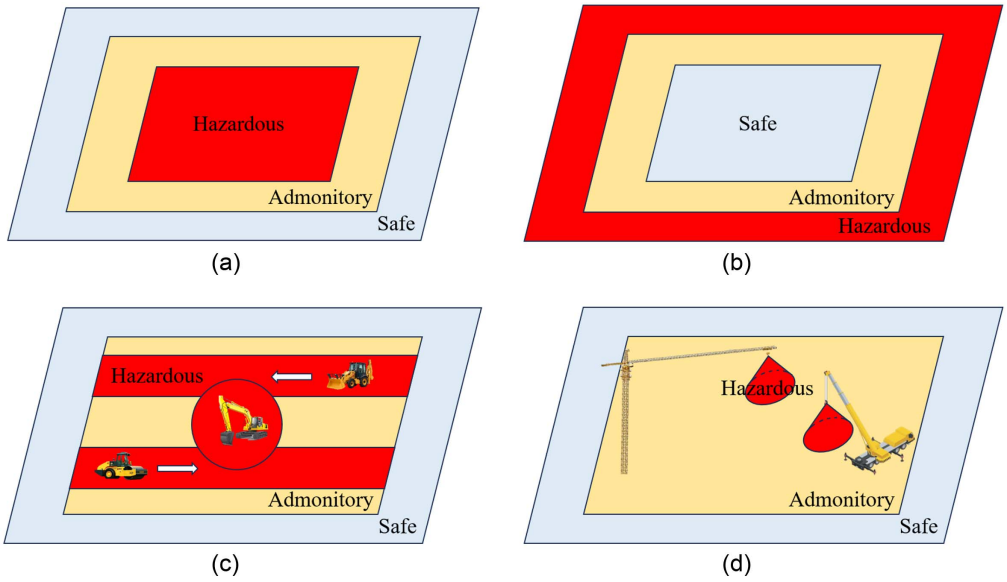


Fig. 2. Classification of near-misses: (a) S1; (b) S2; (c) S3; and (d) S4.

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cameras deployed at the construction site. The image processing module then transmits the processed messages to the digital twin module in real time through the network protocol. In the digital twin module, the construction site is reconstructed in real time, and near-misses can be detected and visualized. Moreover, an early warning module is added to the digital twin model. Managers can guide construction workers based on statistical near-misses, and workers can realize their unsafe behaviors based on visual near-misses.

Physical Module

Multisource Information of Construction Sites

Construction sites often are chaotic and complex scenes in which various elements interact (Haslam et al. 2005), including workers, materials, structures, and machinery (Fig. 4). To ensure workers' safety by monitoring near-misses, it is particularly important to obtain multisource information at the construction sites (Ye et al. 2023). These multisource data, including details about workers, materials, machinery, the building structure itself, and the surrounding environment, form the essential foundation for the digital representation and simulation in our digital twin approach. By capturing and integrating these complex interactions and dynamics within the construction site into the digital model, we are able to conduct in-depth analysis and make informed decisions to enhance safety and optimize construction processes.

Data Acquisition and Image Processing Module

To effectively obtain information at the construction sites in real time, drones and binocular cameras are used to obtain information on static elements and dynamic elements of the construction site respectively. Uncrewed aerial vehicles (UAVs) and surveillance cameras deployed at the construction sites transmit the collected image data to the image processing model in real time for image processing (Fig. 5). The workflow of the model is as follows: the images collected by the drone that are passed into the image processing model are used directly to establish the building information modeling (BIM) model of the construction site, whereas the left and right images collected by the binocular camera are used for target recognition, target tracking, and stereo matching.

The target recognition algorithm is based on Yolov5, which is efficient and effective in identifying workers, materials, and construction machinery in each frame of image (Tan et al. 2021; Zhou et al. 2022). Yolov5 utilizes a single neural network to predict bounding boxes and class probabilities directly from full images in one evaluation, making it highly efficient and capable of real-time processing. Its architecture, which includes a backbone for feature extraction, a neck for feature aggregation, and a head for prediction, ensures high accuracy and robustness in complex environments.

The identified targets are tracked based on Deep SORT version 1.2 that has the advantages of high real-time performance, high accuracy, strong robustness, and strong scalability (Maung and Myint 2022). Deep SORT extends the traditional sort algorithm by incorporating appearance information and deep learning features to handle occlusions and track identity switches more effectively. This integration enhances the tracker's ability to maintain consistent identities across frames, even in crowded or dynamic scenes.

To obtain the three-dimensional world coordinates of the target, stereo calibration, stereo rectification, and stereo matching are required. This paper utilizes (Zhang's 1999) calibration method for the visual localization system. Zhang's calibration method is a widely used technique for camera calibration due to its simplicity and accuracy, allowing for the precise determination of camera parameters from a set of images of a flat pattern.

Subsequently, the internal and external camera parameters are imported seamlessly into OpenCV version 4.7.0. A stereo matching algorithm, which was developed based on semiglobal matching (SGM) (Hirschmuller 2008), is employed for the stereo matching step. SGM is a robust algorithm that computes disparity maps by minimizing a global energy function, and is particularly effective in handling occlusions and textureless regions. This method ensures accurate and dense disparity maps, which is crucial for precise 3D reconstruction.

Ultimately, the three-dimensional information is computed using the principles of similar triangles. Static-element messages and dynamic-element messages then are passed to the digital twin module for construction site reconstruction.

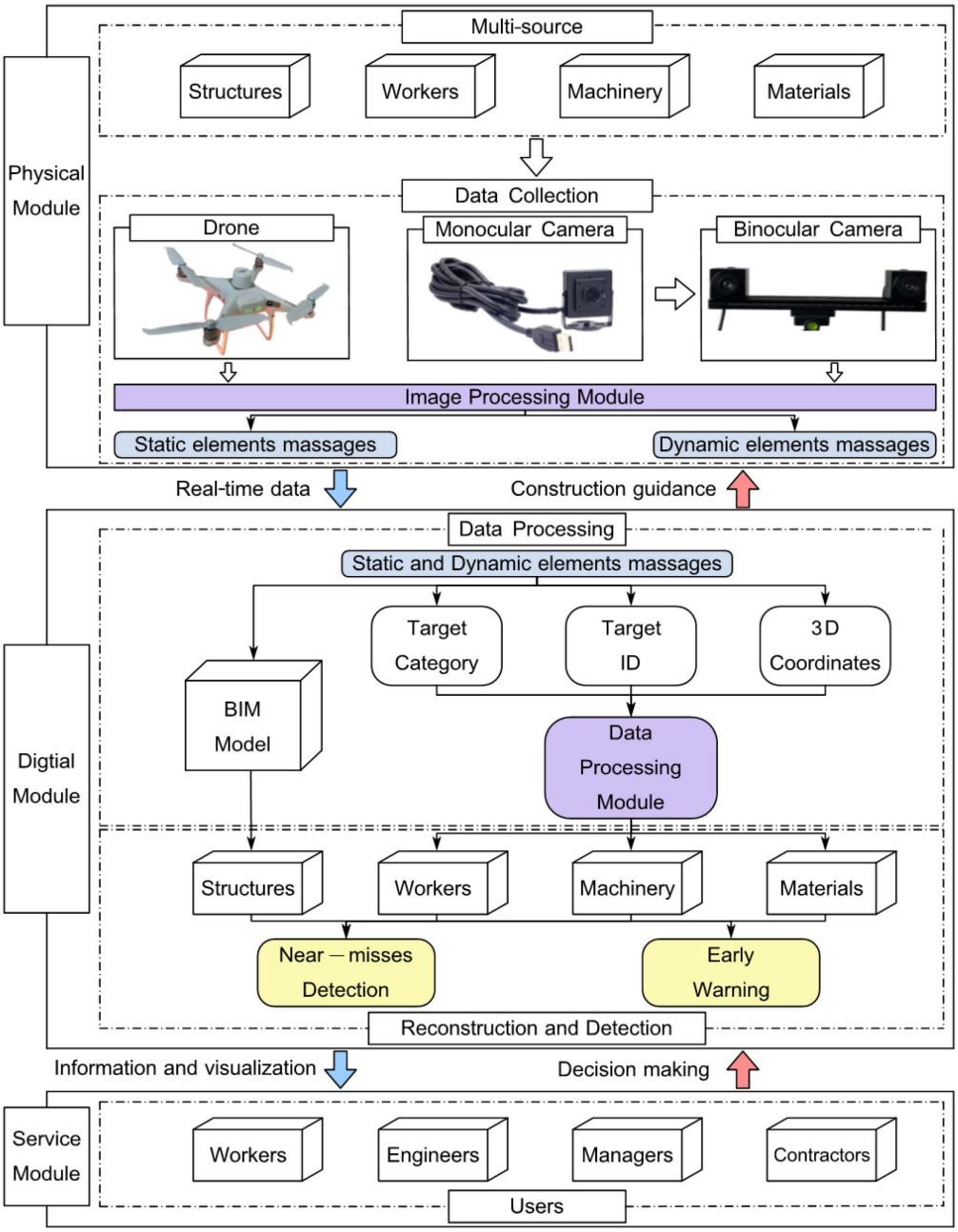


Fig. 3. Framework of the system. (Images by authors.)

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Principle of Binocular Camera Ranging

To enhance the accuracy of locating the target in three-dimensional space, binocular cameras are used. Typically, a binocular stereo vision sensor (Liu and Chen 2009; Yang et al. 2018) comprises a pair of cameras with identical specifications, including focal length, aperture, and sensor dimensions. For optimal results, both the left and right cameras should be aligned on the same plane. This alignment ensures that the horizontal axes of each camera system are collinear and parallel to the image plane (Fig. 6).

In Fig. 6, $P(X, Y, Z)$ as the point of interest, and $P_l(u_l, v_l)$ and $P_r(u_r, v_r)$ are the corresponding projections of this point within the imaging planes of the left camera $O_l x_l y_l$, and the right camera $O_r x_r y_r$, respectively. In addition, B is the baseline, which is the distance separating the left and right cameras. The three-dimensional coordinates are mapped onto an ideal two-dimensional coordinate plane (Fig. 7).

In Fig. 7, f denotes the cameras' focal length, and Z is the depth, which is the distance from Target point P to the binocular camera system. Employing the principle of similar triangles allows us to deduce the following relationship:

$$\frac{B-(u_l-u_r)}{B}=\frac{Z-f}{Z} \tag{1}$$

where $u_l - u_r$ = parallax value. Letting $d = u_l - u_r$, Eq. (1) can be reformulated as follows:

$$\frac{B-d}{B}=\frac{Z-f}{Z} \tag{2}$$

which can be rewritten

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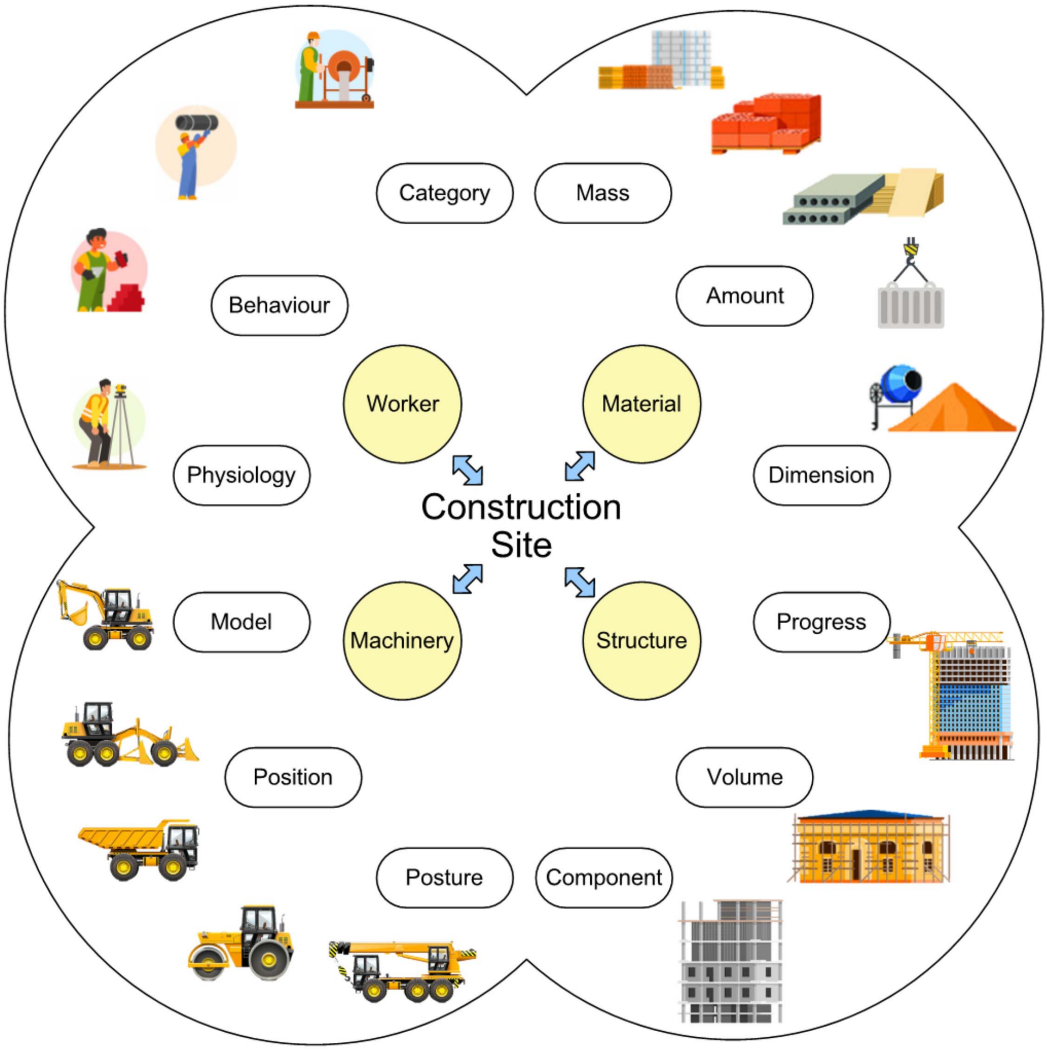


Fig. 4. Multisource data from construction sites. (Construction vehicle icons from Freepik.com/dashadima.)

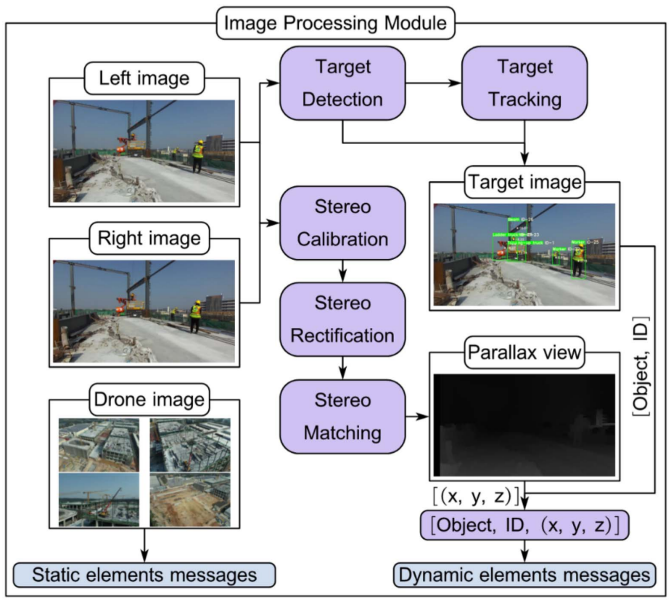


Fig. 5. Image processing module.

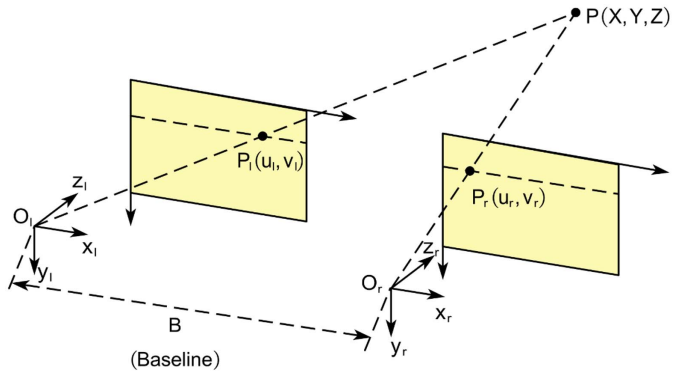


Fig. 6. Principle of binocular vision.

$$Z = f \times \frac{B}{d} \tag{3}$$

From Eq. (3), it is clear that to determine the target's distance from the camera, denoted Z, one must know both the internal and external parameters of the camera system and the parallax

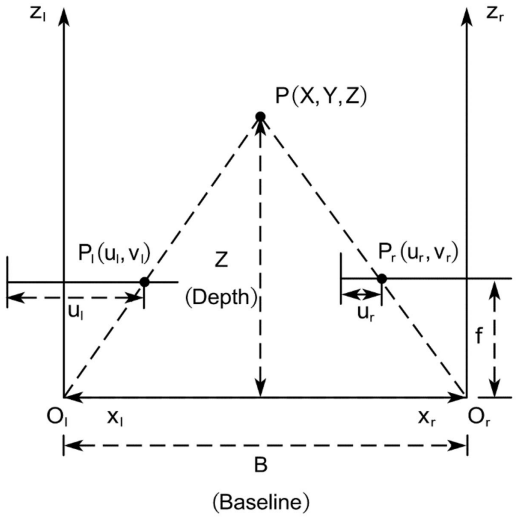


Fig. 7. Ideal two-dimensional plane of the binocular vision principle.

map. However, in a practical camera system, lens aberrations are inevitable, and the imaging planes of the left and right cameras may not align precisely with those of the ideal model. Consequently, it is essential to calibrate the binocular camera system to obtain accurate matrices for the camera's internal and external parameters.

Digital Module

Construction Site Digital Twin Model

The construction site virtual model is built on Unreal Engine version 5.1.1 (Qiu and Yuille 2016) based on static element messages and dynamic element messages. The BIM model built based on static element information is imported into Unreal Engine, and surrounding environmental elements are added to model the interactions between building components and environments. The BIM model typically is updated on a daily basis, with dynamic elements being updated in real-time. This frequency generally is adequate for detecting and warning about near-miss events at construction sites. It is acknowledged that different construction sites have unique characteristics, and the update frequency can be adjusted based on the specific project's needs.

The steps to create dynamic elements are shown in Fig. 8. First, a complete dynamic element model database is established, including common workers, materials, and construction machinery at the construction site. The second step is to receive and process the dynamic element information passed into the digital module, which contains all target types, target IDs, and the three-dimensional

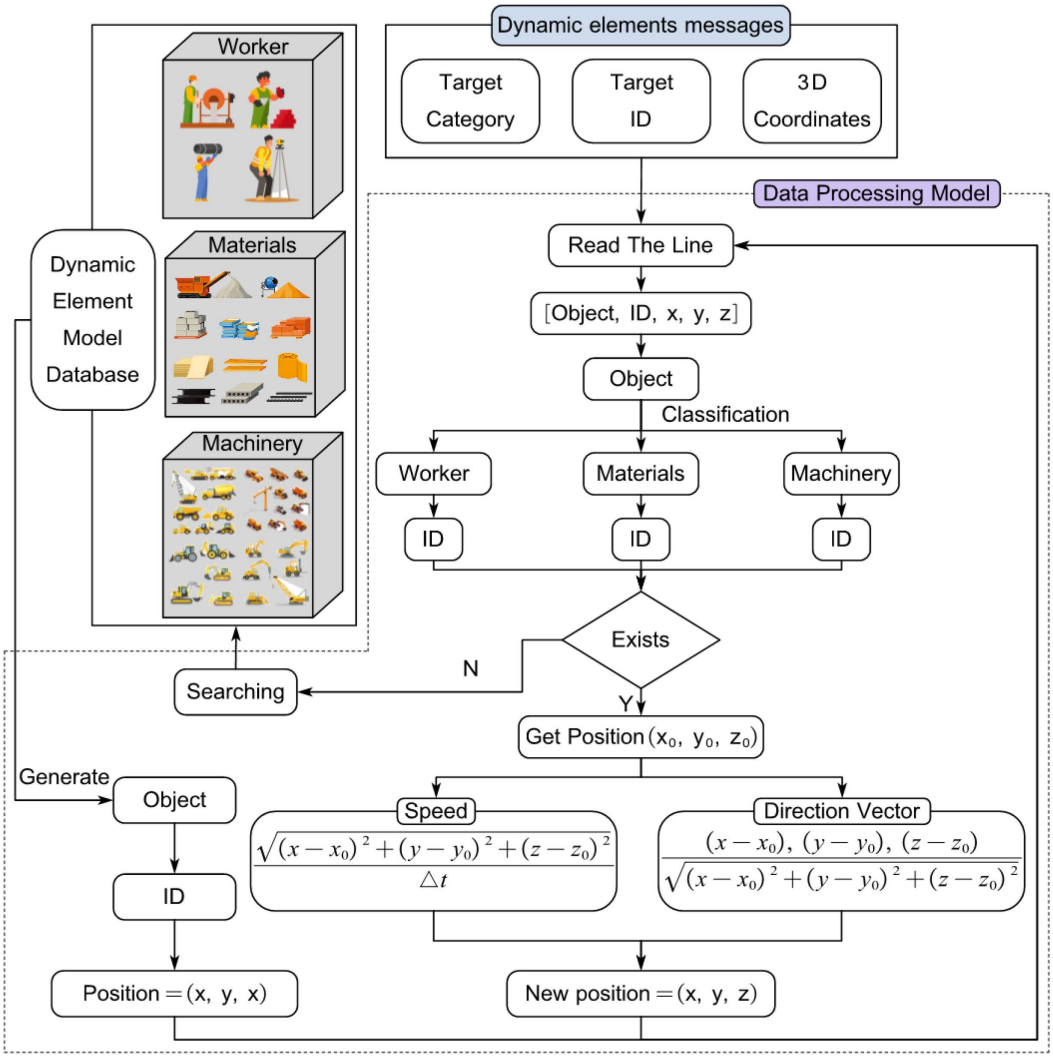


Fig. 8. Generation of dynamic elements.

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world coordinates of the target with the camera as the origin in this frame, that is, multiple lines of information such as [object, ID, x, y, z]. In Unreal Engine, the dynamic element update strategy is implemented using Blueprints to create the data processing module. First, the incoming information is read line by line, and it is determined whether the element already exists based on the target category and its unique ID. If it exists, its traveling speed and direction vector through the coordinates of the starting point and end point and the moving time are calculated, and then it is moved to the target position; if it does not exist, the corresponding model is called in the dynamic element model database and the instance is generated at the target location.

3D Near-Miss Detection Method

To automatically detect and analyze near-misses at construction sites, additional descriptive information for each individual resource involved in a near-miss event is necessary. For example, according to the characteristics of each object, a protective envelope, which users can customize according to different situations, is set to expand a certain range around it. To save computing time, the protective envelope typically is modeled as a cube, sphere, or cylinder in this paper.

Unless specified otherwise by a user, each resource boundary is surrounded by its own protection envelope that remains transparent during operation (Fig. 9). The protection envelopes are determined based on the specific characteristics and operational parameters of the resources involved in the construction site. For the static parts of the construction site, such as structural holes, foundation pits, and unfenced floor edges, the protection envelopes are predefined based on the hazardous nature of these areas. For example, a structural hole might have a protection envelope that extends 2 m beyond its edge to ensure a sufficient safety buffer for workers. For the dynamic parts of the construction site, the protection envelope for workers is set as a cylinder with a radius of 1.5 m, derived from the average distance a person can walk in 1 s, react, and come to a complete stop. The protection envelope for construction materials varies based on their characteristics; for example, large steel beams being lifted by cranes would have larger protection envelopes than a stationary piles of bricks, accounting for their movement and potential risk radius. For construction machinery, several parameters are considered, including operating speed, operating angle, equipment size, and even ground conditions that might affect the machinery's braking distance.

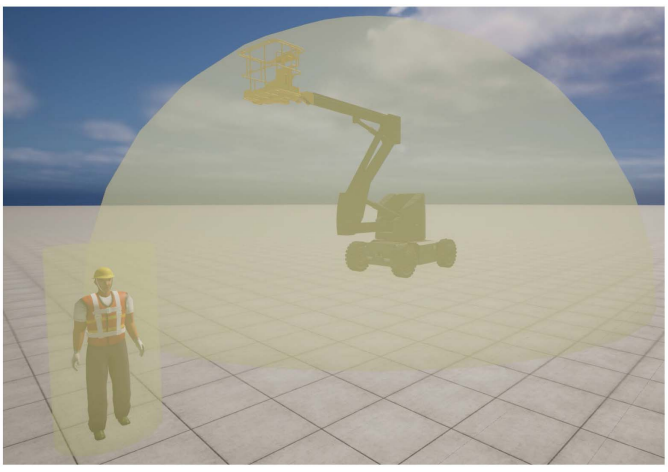


Fig. 9. Protection envelopes of a worker and an excavator.

The current protection envelopes for static and dynamic elements were determined by referring to some existing relevant literature (Speiser and Teizer 2023) and in collaboration with construction site management personnel. The protection envelopes are based on scenarios from the majority of pilot studies, making them applicable to similar construction scenarios. For different projects, users can directly adjust the size of the protection envelopes according to the aforementioned principles by inputting scale-up or scale-down factors, based on the specific characteristics of the project. For example, in a high-rise building project, cranes lifting heavy materials will have a larger protection envelope to ensure safety considering the height and the swing radius of the crane arm. In a road construction project, vehicles such as asphalt pavers and rollers will have protection envelopes adjusted based on their operating speed and the width of the road being constructed. During the construction of a bridge, protection envelopes around the edge of the bridge deck will be significantly extended due to the high risk of falls and the complexity of the operations involved.

Near-misses, as defined in this study, refer to close incidents between one or more pedestrian workers and hazardous objects, resulting in workers being exposed to danger. For example, for two resources A and B, near-misses are defined as collisions of protective envelopes between the two. The near-misses detection algorithm determines their collision start events and end events, and records the timestamps of the collision start and collision end to calculate the duration of the near-misses event.

Service Module

3D Near-Miss Detection Service

The real-time detection of near-misses is displayed on a digital twin interface, and an intelligent risk accumulation system logs these incidents (Fig. 10). This system enables users to modify the threshold to suit various construction project needs. Defaulting to Heinrich's law (Heinrich 1941) and for general applicability, the threshold is preset at 300; surpassing this number prompts the system to alert managers and workers, warning them of the potential for minor injuries. However, users can adjust this threshold according to the specific characteristics and risks associated with different construction activities, allowing for better customization and improved safety management.

Additionally, the analysis of data on near-misses collected from the construction sites is tremendously valuable to both workers and managers. For the workers, these data serve to enhance their awareness and conduct regarding safety. Through the analysis of near-miss incidents, workers can gain a better grasp of potential hazards and proactively take measures to prevent accidents. This not only helps in diminishing the risk of workplace injuries but also elevates the overall safety of the work environment.

For managers, the data on near-misses offer critical insights that can be leveraged to increase the safety of the entire construction project. By examining these data, managers can pinpoint potential security loopholes and areas of concern, enabling them to refine work processes, conduct relevant training, upgrade equipment, or institute new safety protocols. This proactive approach does not only mitigate the risk of accidents, it also contributes to lowering project costs and enhancing productivity.

Proactive Early Warning Service

To pre-empt actual hazardous situations during the near-miss detection process, an early warning mechanism is integrated into the digital twin system. When a near-miss event is identified, the system promptly triggers an alert to caution workers at the construction site about potential dangers, thereby heightening their safety

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Fig. 10. Real-time digital twin–driven near-miss detection system.

consciousness. Concurrently, site managers have access to real-time updates on near-miss incidents and early warnings, enabling them to oversee the entire construction operation more effectively. This real-time monitoring empowers managers to identify anomalies swiftly and implement timely corrective actions.

Pilot Study

To assess the effectiveness of the system introduced in the previous section, an industrial park project located in Foshan City of Guangdong Province in China was selected as a case to demonstrate its implementation performance. The project is composed of six factory buildings, three single-story workshops, and two three-story ramps. This paper focuses on the steel structure hoisting operations on the top floor of the B2 factory building. The structure had not yet been capped, and there are relatively few safety protection measures at the construction site. In addition, during the construction process, workers exhibited many irregular behaviors, which created numerous safety hazards during the steel beam hoisting operations. This case study aimed to improve the safety management of the construction site and provide a strong guarantee for the smooth progress of the project through real-time monitoring of near-misses during steel beam hoisting construction and providing real-time safety warnings.

System Deployment and Equipment Installation

Images of the B2 building were collected using drones, and a BIM model of the construction site was established in advance based on the collected image information to construct our digital twin system. It was imported into Unreal Engine and the surrounding environment was established (Fig. 11). Appropriate protective envelopes in the dangerous areas of the static part, which in this case were floor edges and floor openings, were established. Subsequently, the binocular camera underwent calibration using the method proposed by Zhang (1999). The calibration process employed a tessellated calibration plate with dimensions 340×260 mm and an accuracy of ± 0.01 mm. To calibrate the camera, images were captured at varying positions of the calibration plate. From these images, the pixel coordinates of the calibration plate's corner points were extracted. These coordinates enabled the computation of the camera's initial internal and external parameters through matrix operations. Distortion coefficients then were

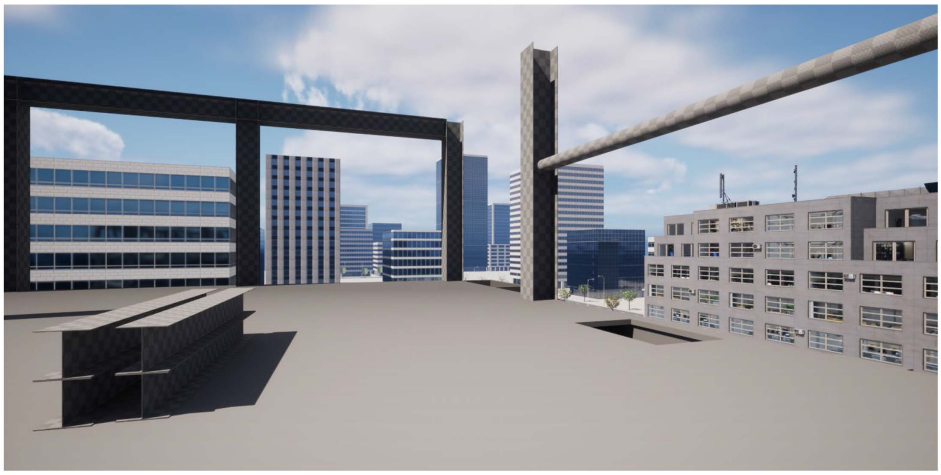


Fig. 11. Digital twin of the construction site.

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Table 1. Internal and external parameters of binocular camera

Parameter	Value
Internal parameters	
Left camera	
Internal parameter matrix	$\begin{bmatrix} 1{,}070 & 0 & 941 \\ 0 & 1{,}070 & 522 \\ 0 & 0 & 1 \end{bmatrix}$
Distortion coefficient	$[-0.0464 \quad 0.0161 \quad 0.0001 \quad -0.0002 \quad 0]$
Right camera	
Internal parameter matrix	$\begin{bmatrix} 1{,}070 & 0 & 941 \\ 0 & 1{,}070 & 522 \\ 0 & 0 & 1 \end{bmatrix}$
Distortion coefficient	$[-0.0446 \quad 0.0134 \quad 0.0001 \quad -0.0002 \quad 0]$
External parameters	
Rotation matrix	$\begin{bmatrix} 0.9998 & -0.0012 & 0.0044 \\ 0.0012 & 0.9998 & -0.0026 \\ -0.0044 & 0.0026 & 0.9998 \end{bmatrix}$
Translation vector	$[119.80 \quad -0.2066 \quad 0.0731]$

estimated using nonlinear least squares, and parameter optimization was accomplished using maximum likelihood estimation. The calibration results are presented in Table 1. Finally, the binocular camera was installed at the construction site to collect dynamic images in real time, with its field of view (FOV) encompassing the steel beam hoisting area.

Application Effects

In this assessment, the defined near-misses were categorized as follows: Type S1 involved floor holes [Fig. 12(a)]; Type S2 corresponded to floor edges lacking fencing [Fig. 12(b)]; Type S3 encompassed the areas surrounding construction machinery [Fig. 12(c)]; and Type S4 related to the space under raised steel beams [Fig. 12(d)].

The digital twin–driven near-miss detection system was applied to construction site hoisting projects; the physical module and digital module are shown in Fig. 13. Twelve workers were engaged in construction tasks throughout the experiment as the system kept vigilant watch over the entire steel beam lifting operation. The ground truth of near-miss events and their quantity in the pilot study were determined by on-site management personnel through manual counting. During the pilot study evaluation, we verified each frame to determine whether a near-miss event occurred in the current frame and whether the digital twin system accurately detected it. Construction site safety management personnel collaborated in the manual counting process to verify the evaluation results of the digital twin system. Within the course of 1 month, there were 1,300 near-miss events. The system not only precisely detected and logged 1,181 of those occurrences, but its early warning system also reliably alerted the workers of the imminent dangers in real time with each near-miss identified.

Based on the results of field tests, a comprehensive evaluation of the model's performance was performed. The precision and recall rates of target detection [Eqs. (4) and (5) (Ke et al. 2016)] were used to measure the near-miss detection accuracy of the model (Table 2). The proposed system successfully identified and detected the four types of near-misses in practice

$$\text{Precision} = \frac{N_{\text{correct}}}{N_{\text{correct}} + N_{\text{incorrect}}} \tag{4}$$

$$\text{Recall} = \frac{N_{\text{correct}}}{N_{\text{correct}} + N_{\text{undetected}}} \tag{5}$$

where N_{correct} = number of near-misses accurately detected by the digital twin; $N_{\text{incorrect}}$ = number of near-misses inaccurately detected; and $N_{\text{undetected}}$ = number of actual near-misses that were not detected by the digital twin system.

Overexposure and underexposure frequently occur at construction sites, necessitating tests of the system's stability across various lighting conditions. The videos saved in the field test were

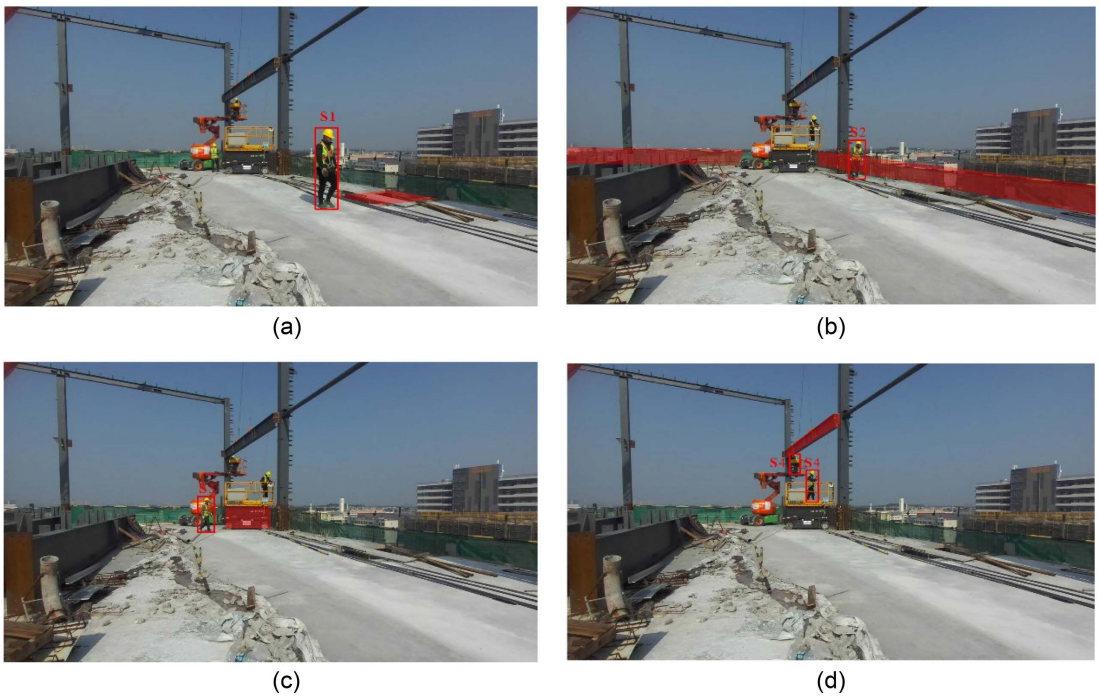


Fig. 12. Near-misses at construction site in pilot study: (a) S1; (b) S2; (c) S3; and (d) S4.

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(a) (b)

Fig. 13. Near-miss detection system: (a) physical module; and (b) digital module.

Table 2. Results of near-miss detection

Performance metric	S1	S2	S3	S4	Total
Correctly detected	237	198	169	577	1,181
Misdetected	0	11	23	44	78
Not detected	0	1	25	15	41
Precision (%)	100.00	94.74	88.02	92.91	93.80
Recall (%)	100.00	99.50	87.11	97.47	96.64

processed for underexposure and overexposure (Fig. 14). Offline testing was performed on the two groups of processed videos. The test results are presented in Table 3 (underexposure) and Table 4 (overexposure). The comparison curves of the three groups of tests are shown in Fig. 15.

The proposed system maintained a precision rate above 90% and a recall rate above 95% across various exposure scenarios, suggesting robust system stability (Tables 3 and 4 and Fig. 15). Precision and recall rates marginally decreased with overexposure compared with underexposure and normal conditions, for which they remained relatively consistent. However, there was a slight variance in precision and recall for different types of near-miss incidents under these exposure conditions. Specifically, the precision for Type S3 near-misses significantly decreased with overexposure. Type S3 consistently had a lower recall rate than Types S1, S2, and S4 across all three exposure scenarios. Moreover, because the early warning system is designed to trigger an alarm upon detection of near-misses, the outcomes from both the early warning and the near-misses detection systems were congruent.

In conclusion, the digital twin–driven near-miss detection system showcased in the study is highly accurate in identifying and detecting near-miss incidents at construction sites, demonstrating commendable stability and minimal susceptibility to varying lighting conditions. Additionally, the integrated early warning system is adept at recognizing hazardous scenarios in real time, ensuring the provision of timely alerts to mitigate potential dangers.

Discussion

This study introduces an image-based digital twin system that, compared with traditional point cloud–based counterparts, demonstrates significant enhancements in real-time capability and lightweight processing. The system has been trained to recognize and track common elements found on construction sites, such as personnel, materials, and machinery, and has established a dynamic element model database. This foundational training ensures that the system is broadly applicable to a wide range of construction environments. However, it is acknowledged that certain specialized equipment and materials may require additional training before deployment. This pretraining process is designed to be efficient, and does not entail an extensive time commitment, allowing for rapid adaptation to specific site requirements. The system's modular design and the use of advanced stereovision techniques facilitate this scalability, making it a robust and flexible tool for enhancing safety management across diverse construction projects.

During field trials, the real-time digital twin–driven system for detecting near-misses demonstrated exceptional performance, achieving a perfect score in both precision and recall for Type S1 near-miss detection. This high level of accuracy may be attributed to the relatively uncluttered environment surrounding the Type S1 near-misses during testing, which were characterized by a lower number of workers and less visual confusion. Additionally, the proximity of Type S1 near-misses to the installed binocular camera, resulting in clearer video imagery, likely contributed to the system's enhanced detection capabilities.

Additionally, the impact of lighting on computer vision is a pivotal factor to take into account, because it greatly influences the visual quality of captured images. Variations in lighting conditions



(a) (b)

Fig. 14. Controlled experiments: (a) underexposure; and (b) overexposure.

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Table 3. Results for underexposure condition

Performance metric	S1	S2	S3	S4	Total
Correctly detected	237	198	169	578	1,182
Misdetected	0	14	9	55	78
Not detected	0	1	25	14	40
Precision (%)	100.00	93.40	94.94	91.31	93.81
Recall (%)	100.00	99.50	87.11	97.64	96.73

Table 4. Results for overexposure condition

Performance metric	S1	S2	S3	S4	Total
Correctly detected	237	195	162	576	1,170
Misdetected	0	8	57	13	78
Not detected	0	4	32	16	52
Precision (%)	100.00	96.06	73.97	97.79	93.75
Recall (%)	100.00	97.99	83.51	97.30	95.74

can lead to pronounced alterations in image attributes, including brightness, contrast, shadows, reflections, and color saturation. Inadequate management of these variations can disrupt the function of image processing and analysis algorithms, leading to compromised performance in critical tasks such as object recognition, tracking, classification, and scene interpretation.

This paper presents an analysis of the system's stability under varying exposure conditions through experimental evaluation. The noticeable decrease in precision for Type S3 near-misses in overexposure scenarios can be attributed to several factors:

- image degradation: overexposed areas within the image can become excessively bright, resulting in a loss of detail;
- feature distortion: overexposure often leads to prominent highlight areas that can overshadow critical features such as edges, shapes, textures, and colors; and
- Data set limitations: the model was trained on a data set lacking diverse exposure scenarios, which hinders its ability to accurately recognize and adapt to real-world exposure variations.

To mitigate these issues, the following methodologies could be considered for future improvements:

- image preprocessing: implementing automatic exposure correction as part of the preprocessing stage can diminish the influence of inconsistent lighting on the system's performance;
- dynamic range extension: employing image sensors with a superior dynamic range can help alleviate the issues of overexposure and underexposure, thus preserving more detail in the imagery; and
- data augmentation: enhancing the training data set to include a wider array of images with varying exposure levels can enable the model to more effectively learn features across different lighting conditions.

The lower recall rates for Type S3 near-misses observed across the three test sets, compared with other types, may be attributable to the densely populated construction machinery present around the S3 areas within the site. The ensuing chaotic and complex environment could result in certain S3 near-misses being undetected by the system, thus contributing to a marginally reduced recall relative to other incident types.

Conclusions and Future Work

The construction industry is characterized by its complexity and variability, leading to a high frequency of safety incidents. To mitigate these risks, it is crucial to implement a real-time monitoring of near-misses at construction sites. Although these near-misses may not result in immediate serious harm, they serve as indicators of underlying risks and vulnerabilities that could precipitate severe injuries or damages. Through proactively analysis of these near-misses, construction management teams can pinpoint and rectify unsafe conditions or practices, effectively forestalling the occurrence of actual accidents. Furthermore, this vigilant monitoring process fosters a culture of safety and heightens the safety consciousness among construction workers, contributing to an overall safety climate.

Digital twin technology has gained significant attention as a state-of-the-art simulation tool across various industries. Therefore, we introduced the application of digital twin technology to the field of near-misses monitoring at construction sites by presenting a digital twin–driven near-miss detection system which distills implicit ontological knowledge within engineering realms. This system adeptly detects near-misses within a three-dimensional space, providing early warnings that underpin a scientific approach to safety management at construction sites. The key contributions of this research are as follows:

- The paper classifies near-misses and multisource data at construction sites, and proposes a real-time information acquisition method based on computer vision, enabling the establishment of a digital twin model for the continuous monitoring of near-misses at construction sites.

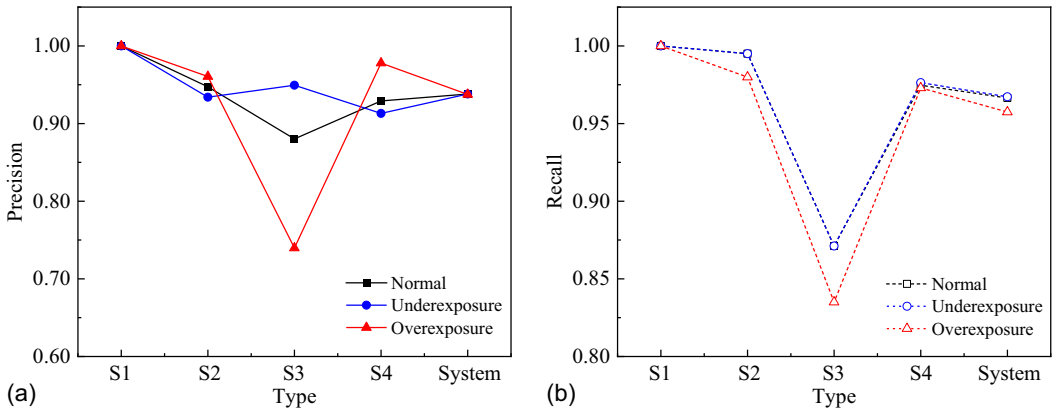


Fig. 15. Effects of underexposure and overexposure: (a) precision; and (b) recall.

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- The paper advances the detection of near-misses at construction sites into a comprehensive three-dimensional domain through the application of digital twin technology. This underscores the proficiency of digital twin technology in tapping into the implicit ontological knowledge pervasive in the engineering sphere, thereby substantially enhancing the efficacy of safety management protocols at construction sites.
- The paper demonstrated the capability of representing near-misses in real-time within the digital twin framework, achieving the visualization of such incidents and offering proactive early warning services. This feature empowers construction managers in gaining a more intuitive grasp of potential hazards on a site. Moreover, the real-time simulation of on-site scenarios furnishes construction workers with more engaging and vivid educational materials for safety training.

However, the research in this paper still has some limitations. Constrained by the inherent limitations of computer vision technology, there is an issue of incomplete information capture when personnel and machinery at the construction site overlap. In addition, certain extreme lighting conditions can marginally affect the stability of the digital twin–driven system for detecting near-misses. Furthermore, current vision-based digital twin system operates effectively within a camera field of view of up to 20 m, which is not suitable for residential houses, high-rise buildings, and large classroom buildings.

Future research will address the occlusion problem by integrating multiple perspectives to broaden the system's applicability and enhance its suitability for large-scale complex scenarios. Additionally, enhancements in both hardware and algorithms will be considered to mitigate the impact of lighting conditions, with the overarching goal of continuously refining the system to further the advancement of safety management in the construction industry. Furthermore, future studies will conduct detailed variable analysis of different update frequencies to balance efficiency and cost, enhancing the practical application of DT technology in various construction settings.

Data Availability Statement

Some or all data, models, or code that support the findings of this study are available from the corresponding author upon reasonable request.

Acknowledgments

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